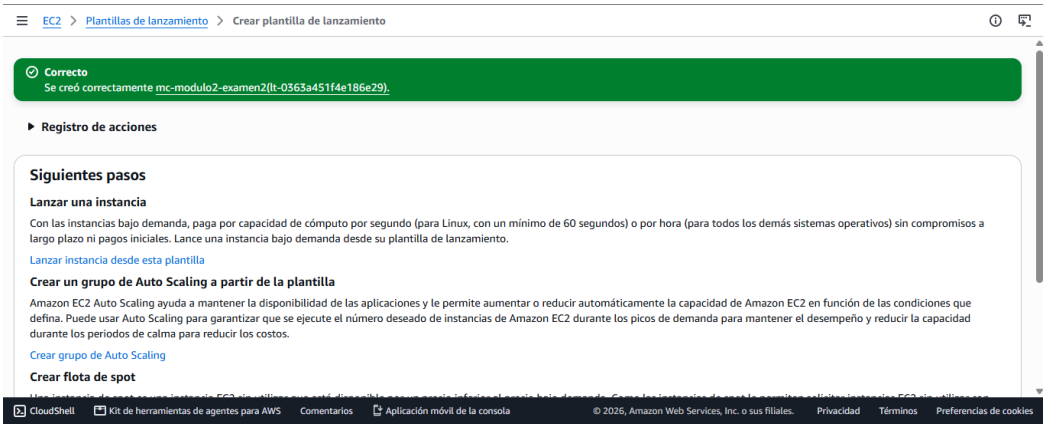
Examen Práctico Módulo 2 – Segunda parte

Computación en la nube – Auto Scaling, CloudWatch, CloudTrail y Lambda

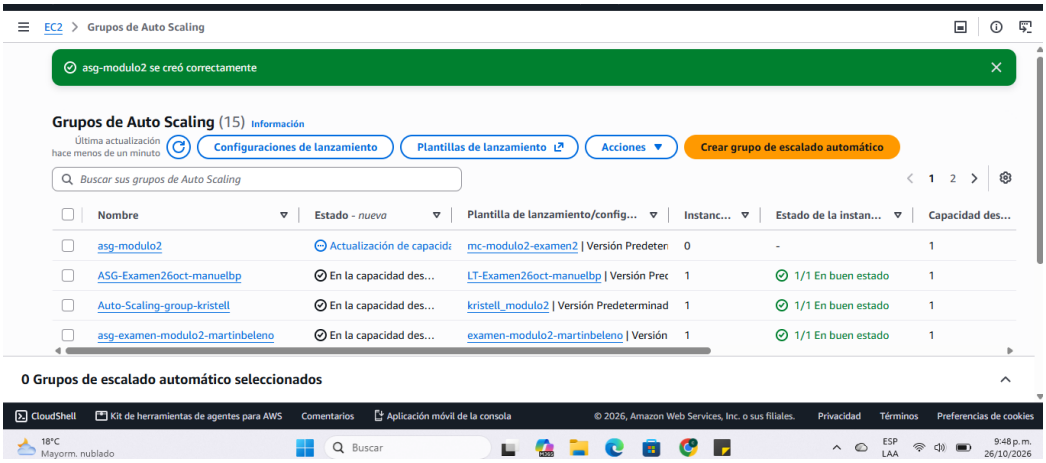
Estudiante	Maria Camila Mejia Arango
Región AWS	us-east-1 (N. Virginia)
Plantilla de lanzamiento	mc-modulo2-examen2 (t2.micro, Amazon Linux 2023)
Auto Scaling Group	asg-modulo2 (deseada 1, mínima 1, máxima 2)
Instancia del grupo	i-0039362d684531802
Función Lambda	lambda-ec2

1. Auto Scaling Group (arquitectura_asg.png)

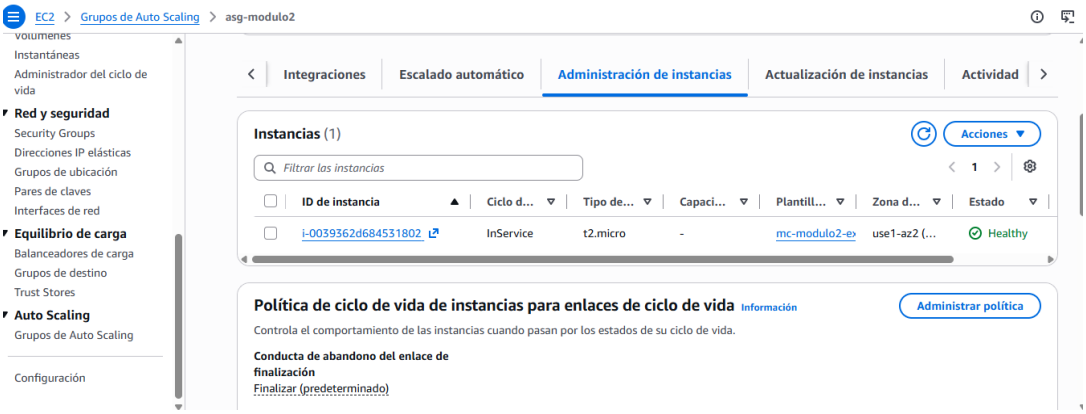
Plantilla de lanzamiento creada



Grupo de Auto Scaling creado



Instancia t2.micro del grupo (InService)



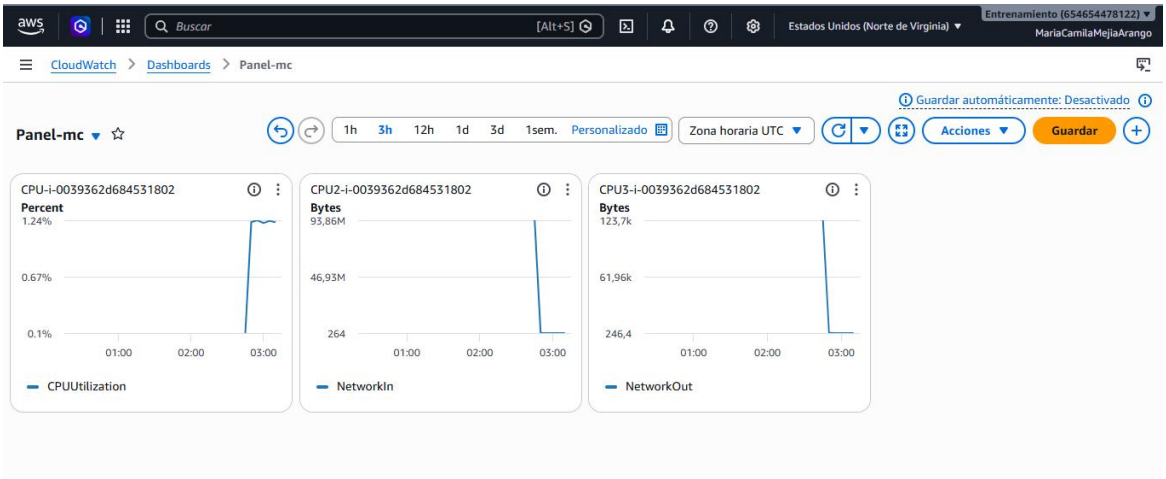
2. Alarmas de CloudWatch

Alarma de advertencia (CPU ≥ 60 %) y alarma crítica (CPU ≥ 90 %).



3. Métricas de la instancia (metricas_instancia.png)

Panel con tres gráficos (CPUUtilization, NetworkIn y NetworkOut) de la instancia i-0039362d684531802.



4. CloudTrail

evento_creacion.json (evento RunInstances)


```
{
    "eventVersion": "1.11",
    "userIdentity": {
        "type": "AssumedRole",
        "principalId": "AROAZQ3DSS4VILINXU7ZK:AutoScaling",
        "arn": "arn:aws:sts::654654478122:assumed-
role/AWSServiceRoleForAutoScaling/AutoScaling",
        "accountId": "654654478122",
        "accessKeyId": "ASIAZQ3DSS4VESL2BYZ6",
        "sessionContext": {
            "sessionIssuer": {
                "type": "Role",
                "principalId": "AROAZQ3DSS4VILINXU7ZK",
                "arn": "arn:aws:iam::654654478122:role/aws-service-
role/autoscaling.amazonaws.com/AWSServiceRoleForAutoScaling",
                "accountId": "654654478122",
                "userName": "AWSServiceRoleForAutoScaling"
            },
            "attributes": {
                "creationDate": "2026-10-27T03:23:06Z",
                "mfaAuthenticated": "false"
            }
        },
        "invokedBy": "autoscaling.amazonaws.com"
    },
    "eventTime": "2026-10-27T03:23:08Z",
    "eventSource": "ec2.amazonaws.com",
    "eventName": "RunInstances",
    "awsRegion": "us-east-1",
    "sourceIPAddress": "autoscaling.amazonaws.com",
    "userAgent": "autoscaling.amazonaws.com",
    "requestParameters": {
        "instancesSet": {
            "items": [
                {
                    "minCount": 1,
                    "maxCount": 1
                }
            ]
        },
        "blockDeviceMapping": {},
        "availabilityZone": "us-east-1b",
        "monitoring": {
            "enabled": false
        },
        "subnetId": "subnet-0d264823f56ab1bc3",
        "disableApiTermination": false,
        "disableApiStop": false,
        "clientToken": "96b685ed-883e-5f35-e869-13f6947f3aa2",
        "tagSpecificationSet": {
```

```
            "items": [
                {
                    "resourceType": "instance",
                    "tags": [
                        {
                            "key": "aws:autoscaling:groupName",
                            "value": "MiASG-Examen2-Tina"
                        }
                    ]
                }
            ]
        },
        "launchTemplate": {
            "launchTemplateId": "lt-0ecc153be00d68229",
            "version": "1"
        }
    },
    "responseElements": {
        "requestId": "648dddc2-d382-41d2-91e9-ef93ba67b7cd",
        "reservationId": "r-09bf4b0a7395641c1",
        "ownerId": "654654478122",
        "groupSet": {},
        "instancesSet": {
            "items": [
                {
                    "instanceId": "i-08223881e8867cfc5",
                    "imageId": "ami-04b938437b69a2c9c",
                    "bootMode": "uefi-preferred",
                    "currentInstanceBootMode": "legacy-bios",
                    "instanceState": {
                        "code": 0,
                        "name": "pending"
                    },
                    "privateDnsName": "ip-10-0-29-22.ec2.internal",
                    "keyName": "Tina-varela-keypair-workshop1",
                    "operator": {
                        "managed": false
                    },
                    "amiLaunchIndex": 0,
                    "productCodes": {},
                    "instanceType": "t2.micro",
                    "launchTime": 1793071388000,
                    "placement": {
                        "availabilityZone": "us-east-1b",
                        "availabilityZoneId": "use1-az2",
                        "tenancy": "default"
                    },
                    "monitoring": {
                        "state": "disabled"
                    },
                    "subnetId": "subnet-0d264823f56ab1bc3",
                    "vpcId": "vpc-0b3d08f5f4fcee21a",
                    "privateIpAddress": "10.0.29.22",
                    "stateReason": {
```

```
                        "code": "pending",
                        "message": "pending"
                    },
                    "architecture": "x86_64",
                    "rootDeviceType": "ebs",
                    "rootDeviceName": "/dev/xvda",
                    "blockDeviceMapping": {},
                    "virtualizationType": "hvm",
                    "hypervisor": "xen",
                    "tagSet": {
                        "items": [
                            {
                                "key": "aws:autoscaling:groupName",
                                "value": "MiASG-Examen2-Tina"
                            },
                            {
                                "key": "aws:ec2launchtemplate:id",
                                "value": "lt-0ecc153be00d68229"
                            },
                            {
                                "key": "aws:ec2launchtemplate:version",
                                "value": "1"
                            }
                        ]
                    },
                    "clientToken": "96b685ed-883e-5f35-e869-13f6947f3aa2",
                    "groupSet": {
                        "items": [
                            {
                                "groupId": "sg-0ed8031138ecc175f",
                                "groupName": "default"
                            }
                        ]
                    },
                    "sourceDestCheck": true,
                    "networkInterfaceSet": {
                        "items": [
                            {
                                "networkInterfaceId": "eni-
0f058d2679f5b9278",
                                "subnetId": "subnet-0d264823f56ab1bc3",
                                "vpcId": "vpc-0b3d08f5f4fcee21a",
                                "ownerId": "654654478122",
                                "operator": {
                                    "managed": false
                                },
                                "status": "in-use",
                                "macAddress": "12:ff:ce:39:1e:95",
                                "privateIpAddress": "10.0.29.22",
                                "privateDnsName": "ip-10-0-29-
22.ec2.internal",
                                "sourceDestCheck": true,
                                "interfaceType": "interface",
                                "groupSet": {
```

```
                                    "items": [
                                        {
                                            "groupId": "sg-
0ed8031138ecc175f",
                                            "groupName": "default"
                                        }
                                    ]
                                },
                                "attachment": {
                                    "attachmentId": "eni-attach-
0749c021483eb917c",
                                    "deviceIndex": 0,
                                    "networkCardIndex": 0,
                                    "status": "attaching",
                                    "attachTime": 1793071388000,
                                    "deleteOnTermination": true
                                },
                                "privateIpAddressesSet": {
                                    "item": [
                                        {
                                            "privateIpAddress":
"10.0.29.22",
                                            "privateDnsName": "ip-10-0-29-
22.ec2.internal",
                                            "primary": true
                                        }
                                    ]
                                },
                                "ipv6AddressesSet": {},
                                "tagSet": {}
                            }
                        ]
                    },
                    "ebsOptimized": false,
                    "enaSupport": true,
                    "cpuOptions": {
                        "coreCount": 1,
                        "threadsPerCore": 1
                    },
                    "capacityReservationSpecification": {
                        "capacityReservationPreference": "open"
                    },
                    "enclaveOptions": {
                        "enabled": false
                    },
                    "metadataOptions": {
                        "state": "pending",
                        "httpTokens": "required",
                        "httpPutResponseHopLimit": 2,
                        "httpEndpoint": "enabled",
                        "httpProtocolIpv4": "enabled",
                        "httpProtocolIpv6": "disabled",
                        "instanceMetadataTags": "disabled"
                    },
```

```
                    "maintenanceOptions": {
                        "autoRecovery": "default",
                        "rebootMigration": "default"
                    },
                    "privateDnsNameOptions": {
                        "hostnameType": "ip-name",
                        "enableResourceNameDnsARecord": false,
                        "enableResourceNameDnsAAAARecord": false
                    }
                }
            ]
        },
        "requesterId": "661627717883"
    },
    "requestID": "648dddc2-d382-41d2-91e9-ef93ba67b7cd",
    "eventID": "01761a96-f922-4523-bee6-8640404b190a",
    "readOnly": false,
    "resources": [
        {
            "accountId": "654654478122",
            "type": "AWS::EC2::Instance",
            "ARN": "arn:aws:ec2:us-east-1:654654478122:instance/i-
08223881e8867cfc5"
        }
    ],
    "eventType": "AwsApiCall",
    "managementEvent": true,
    "recipientAccountId": "654654478122",
    "eventCategory": "Management"
}
```


analisis_trail.md

¿Desde qué dirección IP o servicio se originó la creación de la instancia según el JSON?

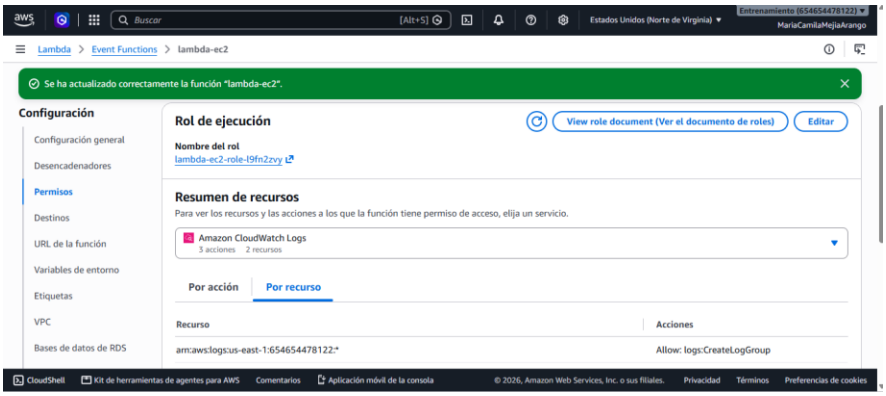
Según el JSON (evento RunInstances), el campo sourceIPAddress es autoscaling.amazonaws.com, por lo que la instancia fue creada por el servicio EC2 Auto Scaling (grupo asg-modulo2) y no desde una IP de usuario.

Últimas 5 acciones del usuario IAM (MariaCamilaMejiaArango)

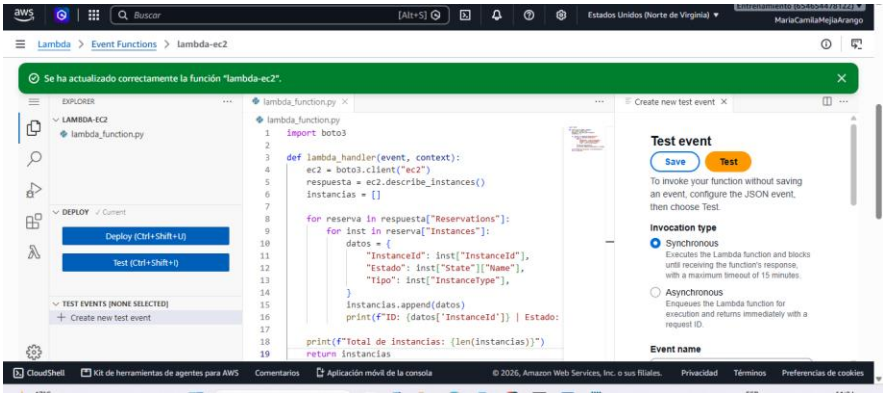
1. ListManagedNotificationEvents
2. ListDiscoveredResources
3. ListDiscoveredResources
4. ListDiscoveredResources
5. ListDiscoveredResources

5. Función Lambda (Lambda.png)

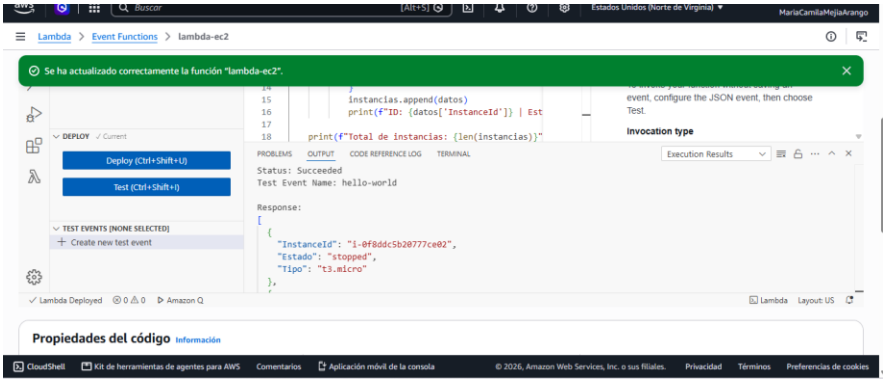
Rol de la función



Código de la función



Ejecución de la prueba



Log group de la Lambda (/aws/lambda/lambda-ec2)

